

Quelle est la différence entre une trame et un paquet ?

une trame est utilisée pour envoyer des données à l'intérieur d'un seul réseau. Un paquet est utilisé pour envoyer des données d'un réseau à un autre, puis à un appareil spécifique sur ce réseau.

Qu'est-ce que le format pcap/pcapng ?

PCAPNG (PCAP Next Generation) est plus extensible et portable que PCAP et possède une structure de fichiers plus flexible. Il inclut également des informations supplémentaires sur le paquet et l'interface utilisée pour sa capture, telles que les compteurs de paquets perdus, les enregistrements DNS, etc.

Quelles sont les adresses MAC sources, les IP sources et les adresses MAC destinations, les IP destinations des données capturées ?

TCP:

84	0.909082200	10.10.15.60	52.113.196.254	TCP	66	63695 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
85	0.921413100	52.113.196.254	10.10.15.60	TCP	66	443 → 63695 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
86	0.921465500	10.10.15.60	52.113.196.254	TCP	54	63695 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0


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▶ Frame 171: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{0...}
▼ Ethernet II, Src: Intel_3a:2e:49 (68:05:ca:3a:2e:49), Dst: Intel_65:aa:4a (04:ec:d8:65:aa:4a)
  ▼ Destination: Intel_65:aa:4a (04:ec:d8:65:aa:4a)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: Intel_3a:2e:49 (68:05:ca:3a:2e:49)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 31]
▶ Internet Protocol Version 4, Src: 150.171.110.49, Dst: 10.10.15.60
▶ Transmission Control Protocol, Src Port: 443, Dst Port: 12039, Seq: 0, Ack: 1, Len: 0
```

arp et udp

183	1.226074300	Intel_13:7b:80	Broadcast	ARP	60	Who has 10.10.15.188? (ARP Probe)
184	1.226074300	QNAP_7e:1f:2b	Broadcast	ARP	60	Who has 10.10.0.228? Tell 10.10.0.200
185	1.226074300	QNAP_7e:1f:2b	Broadcast	ARP	60	Who has 10.10.1.205? Tell 10.10.0.200
186	1.226074300	10.10.14.141	10.10.255.255	UDP	50	2008 → 2008 Len=8

mac udp

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▶ Frame 186: Packet, 50 bytes on wire (400 bits), 50 bytes captured (400 bits) on interface \Device\NPF_{0...}
▼ Ethernet II, Src: Intel_99:4a:4e (bc:38:98:99:4a:4e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... ..1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: Intel_99:4a:4e (bc:38:98:99:4a:4e)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 15]
▶ Internet Protocol Version 4, Src: 10.10.14.141, Dst: 10.10.255.255
▶ User Datagram Protocol, Src Port: 2008, Dst Port: 2008
▶ Data (8 bytes)
```

mac arp

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▶ Frame 185: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{0...}
▼ Ethernet II, Src: QNAP_7e:1f:2b (24:5e:be:7e:1f:2b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... ..1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: QNAP_7e:1f:2b (24:5e:be:7e:1f:2b)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 3]
  Padding: 00000000000000000000000000000000
▶ Address Resolution Protocol (request)
```

autre protocole : mdns

TLSv1.3 et TLSv1.2.

dhcp

sns

Pour le TCP, essayez de trouver les paquets correspondants aux étapes de connexion entre votre hôte et un serveur (intra).

590	1.845028600	138.199.132.59	10.10.18.177	TLSv1.3	1409 Application Data, Application Data
591	1.863676400	10.10.18.177	138.199.132.59	TLSv1.3	161 Application Data
592	1.864630500	10.10.18.177	138.199.132.59	TLSv1.3	160 Application Data
593	1.869697200	138.199.132.59	10.10.18.177	TCP	54 443 → 20374 [ACK] Seq=11376 Ack=5298 Win=81536 Len=0
594	1.893208200	138.199.132.59	10.10.18.177	TCP	54 443 → 20374 [ACK] Seq=11376 Ack=5511 Win=81536 Len=0
595	1.893844800	138.199.132.59	10.10.18.177	TCP	1514 443 → 20374 [ACK] Seq=11376 Ack=5511 Win=81536 Len=1460 [TCP PDU reassembled in 597]
596	1.893844800	138.199.132.59	10.10.18.177	TCP	1514 443 → 20374 [PSH, ACK] Seq=12836 Ack=5511 Win=81536 Len=1460 [TCP PDU reassembled in 597]
597	1.893844800	138.199.132.59	10.10.18.177	TLSv1.3	811 Application Data, Application Data
598	1.893844800	138.199.132.59	10.10.18.177	TLSv1.3	653 Application Data, Application Data
599	1.893962500	10.10.18.177	138.199.132.59	TCP	54 20374 → 443 [ACK] Seq=5511 Ack=15652 Win=65280 Len=0
600	1.911837600	138.199.132.59	10.10.18.177	TLSv1.3	869 Application Data, Application Data
601	1.912639100	10.10.18.177	138.199.132.59	TLSv1.3	5429 Application Data
602	1.912738200	10.10.18.177	138.199.132.59	TLSv1.3	160 Application Data
603	1.914031900	10.10.18.177	138.199.132.59	TLSv1.3	192 Application Data
604	1.916427700	138.199.132.59	10.10.18.177	TLSv1.3	676 Application Data, Application Data
605	1.942407700	138.199.132.59	10.10.18.177	TCP	54 443 → 20374 [ACK] Seq=17089 Ack=9891 Win=81536 Len=0

Documentez-vous sur le sujet et faites quelques tests pour n'afficher que les trames qui vous intéressent.

utiliser la barre de recherche wireshark pour filtrer les protocoles

Partie 2

def mDNS

protocole de résolution de noms sans avoir à configurer un serveur DNS. Il permet aux hôtes de communiquer entre eux dans votre réseau au sein du même VLAN sans interruption. mDNS utilise des paquets multicast pour permettre la découverte des différents hôtes dans le réseau.

filtre : "dns or tcp or http or mdns or ssl or smb or ftp or tls or dhcp"

No.	Time	Source	Destination	Protocol	Length	Info
1782	15.568264000	10.10.19.201	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
1784	15.568264000	10.10.18.160	224.0.0.251	MDNS	380	Standard query response 0x0000 PTR BeyondTV-ce300ec92f2a372935080bf7ac9599de._googlecast._tcp.local TXT, cache flush SRV, cache
1785	15.568264000	10.10.18.160	224.0.0.251	MDNS	380	Standard query response 0x0000 PTR BeyondTV-ce300ec92f2a372935080bf7ac9599de._googlecast._tcp.local TXT, cache flush SRV, cache
1789	15.610032000	10.10.19.215	34.125.0.111	TCP	55	22567 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
1790	15.758967500	34.125.0.111	10.10.19.215	TCP	70	443 → 22567 [ACK] Seq=1 Ack=2 Win=496 Len=0 SLE=1 SRE=2
1801	15.963439400	10.10.19.215	10.10.0.1	DNS	83	Standard query 0x2f47 A www.msftconnecttest.com
1802	15.966056400	10.10.0.1	10.10.19.215	DNS	193	Standard query response 0x2f47 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23
1803	15.966847600	10.10.19.215	23.41.187.33	TCP	66	15412 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1809	15.980944500	10.10.18.18	224.0.0.251	MDNS	107	Standard query 0x0000 TXT Nats's MacBook Air._companion-link._tcp.local, "QU" question
1818	15.987881200	23.41.187.33	10.10.19.215	TCP	66	80 → 15412 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1819	15.987973600	10.10.19.215	23.41.187.33	TCP	54	15412 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0 MSS=1460 SACK_PERM WS=128
1820	15.988153300	10.10.19.215	23.41.187.33	HTTP	165	GET /connecttest.txt HTTP/1.1
1821	16.009350000	23.41.187.33	10.10.19.215	TCP	54	80 → 15412 [ACK] Seq=1 Ack=112 Win=64256 Len=0
1822	16.010713800	23.41.187.33	10.10.19.215	HTTP	241	HTTP/1.1 200 OK (text/plain)
1823	16.010713800	23.41.187.33	10.10.19.215	TCP	54	80 → 15412 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
1824	16.010800900	10.10.19.215	23.41.187.33	TCP	54	15412 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
1825	16.010919100	10.10.19.215	23.41.187.33	TCP	54	15412 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
1826	16.031933000	23.41.187.33	10.10.19.215	TCP	54	80 → 15412 [ACK] Seq=189 Ack=113 Win=64256 Len=0
1831	16.186647400	10.10.19.149	224.0.0.251	MDNS	184	Standard query 0x0004 PTR _googlecast._tcp.local, "QM" question PTR _X9E5E7C8F47989526C9BCD95D24084F6F0827C5ED._sub._googlecast._

En écoutant des échanges FTP sans TLS, que remarquez-vous dans les paquets ? Est-il possible de récupérer des données sensibles de connexion ?

54	10.981065000	192.168.35.1	192.168.35.166	FTP	77 Request: CWD /home/evan/Images
55	10.981768100	192.168.35.166	192.168.35.1	FTP	96 Response: 250 Commande CWD exécutée avec succès
56	10.981928800	192.168.35.1	192.168.35.166	FTP	59 Request: PWD
57	10.982297000	192.168.35.166	192.168.35.1	FTP	106 Response: 257 "/home/evan/Images" est le répertoire courant
58	10.982422500	192.168.35.1	192.168.35.166	FTP	60 Request: PASV
59	10.982849200	192.168.35.166	192.168.35.1	FTP	105 Response: 227 Entering Passive Mode (192,168,35,166,168,5).
60	10.982975500	192.168.35.1	192.168.35.166	FTP	60 Request: MLSD
61	10.983097200	192.168.35.1	192.168.35.166	TCP	66 3195 → 43013 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=128 SACK_PERM
62	10.983210800	192.168.35.166	192.168.35.1	TCP	66 43013 → 3195 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
63	10.983303500	192.168.35.1	192.168.35.166	TCP	54 3195 → 43013 [ACK] Seq=1 Ack=1 Win=4194304 Len=0
64	10.983525500	192.168.35.166	192.168.35.1	FTP	122 Response: 150 Ouverture d'une connexion de données en mode BINARY pour MLSD
65	10.983775700	192.168.35.166	192.168.35.1	FTP-Data...	526 FTP Data: 472 bytes (PASV) (MLSD)
66	10.983853200	192.168.35.1	192.168.35.166	TCP	54 3195 → 43013 [ACK] Seq=1 Ack=474 Win=4193792 Len=0

les données ne sont pas cryptées en FTP

oui, il est possible de recueillir des données sensibles car les données s'affichent en clair

En est-il de même avec les échanges SSL ?

non car les données sont cryptées

connecter en SFTP

37	70.295265500	192.168.35.166	192.168.35.1	TCP	66 22 → 53859 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
38	70.295398200	192.168.35.1	192.168.35.166	TCP	54 53859 → 22 [ACK] Seq=1 Ack=1 Win=4194304 Len=0
39	70.295504800	192.168.35.1	192.168.35.166	SSHv2	80 Client: Protocol (SSH-2.0-FileZilla_3.69.6)
40	70.295624000	192.168.35.166	192.168.35.1	TCP	60 22 → 53859 [ACK] Seq=1 Ack=27 Win=64256 Len=0
41	70.301925100	192.168.35.166	192.168.35.1	SSHv2	95 Server: Protocol (SSH-2.0-OpenSSH_10.0p2 Debian-7+deb13u2)
42	70.302953700	192.168.35.1	192.168.35.166	SSHv2	1430 Client: Key Exchange Init
43	70.307813000	192.168.35.166	192.168.35.1	SSHv2	1094 Server: Key Exchange Init
44	70.309101500	192.168.35.1	192.168.35.166	SSHv2	102 Client: Elliptic Curve Diffie-Hellman Key Exchange Init
45	70.311842500	192.168.35.166	192.168.35.1	SSHv2	554 Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=292)
46	70.317088900	192.168.35.1	192.168.35.166	SSHv2	122 Client: New Keys, Encrypted packet (len=52)
47	70.317339700	192.168.35.166	192.168.35.1	SSHv2	106 Server: Encrypted packet (len=52)
48	70.319175800	192.168.35.1	192.168.35.166	SSHv2	122 Client: Encrypted packet (len=52)
49	70.320041900	192.168.35.166	192.168.35.1	SSHv2	106 Server: Encrypted packet (len=52)
50	70.320268100	192.168.35.1	192.168.35.166	SSHv2	318 Client: Encrypted packet (len=264)
51	70.361878900	192.168.35.166	192.168.35.1	TCP	60 22 → 53859 [ACK] Seq=1686 Ack=1851 Win=70016 Len=0
52	70.431642300	192.168.35.166	192.168.35.1	SSHv2	90 Server: Encrypted packet (len=36)
53	70.432583400	192.168.35.1	192.168.35.166	SSHv2	106 Client: Encrypted packet (len=52)
54	70.432821100	192.168.35.166	192.168.35.1	TCP	60 22 → 53859 [ACK] Seq=1722 Ack=1903 Win=70016 Len=0
55	70.468830400	192.168.35.166	192.168.35.1	SSHv2	682 Server: Encrypted packet (len=628)

> Frame 48: Packet, 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface \Device\NPF_{...}	0000 00 0c 29 24 46 62 00 50 56 c0 00 08 08 00 45 00 ...)\$Fb P VE
> Ethernet II, Src: VMware_c0:00:08 (00:50:56:c0:00:08), Dst: VMware_24:46:62 (00:0c:29:24:46:62)	0010 00 6c 74 76 40 00 80 06 be 1d c0 a8 23 01 c0 a8 ... ltv@.....#...
> Internet Protocol Version 4, Src: 192.168.35.1, Dst: 192.168.35.166	0020 23 a6 d2 63 00 16 a7 60 c9 8c bc 20 8b 4c 50 18 ... #.....LP
> Transmission Control Protocol, Src Port: 53859, Dst Port: 22, Seq: 1519, Ack: 1634, Len: 68	0030 7f ff 71 fd 00 00 00 00 00 30 28 6b 67 63 69 ce ... q.....0(k ci
> SSH Protocol	0040 6e 40 6e 30 1d 79 4c 95 fe 1b ad ba 3f 75 33 45 ... n-n8 yL... 2u3E
	0050 ce 34 1e 81 49 8a f8 37 2a 1d ce 13 99 4d 20 24 ... 4 I 7 *... M \$
	0060 34 5b 2a c6 37 47 29 46 87 4f 9a ef 48 8e b4 de ... 4[* 7G)F 0 H...
	0070 f6 1a 78 23 81 41 3b 15 e3 41 ... x# A; ... A

Partie 3

commande qui vous permettra

d'écouter et de capturer les paquets de quelques-uns de protocoles listés.

Filtre d'affichage

“tshark -i ens33 “tcp” : permet de capturer uniquement le protocole tcp

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root@debian:~# tshark -i ens33 "tcp"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'ens33'
  1 0.000000000 192.168.35.166 -> 34.160.144.191 TCP 74 36346 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=521529980 TSecr=0 WS=128
  2 0.010681857 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
  3 0.010723880 192.168.35.166 -> 34.160.144.191 TCP 54 36346 -> 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
  4 0.010928034 192.168.35.166 -> 34.160.144.191 TLSv1 274 Client Hello (SNI=content-signature-2.cdn.mozilla.net)
  5 0.011156491 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=1 Ack=221 Win=64240 Len=0
  6 0.039042713 34.160.144.191 -> 192.168.35.166 TLSv1.2 3074 Server Hello, Certificate, Server Key Exchange, Server Hello Done
  7 0.039077202 192.168.35.166 -> 34.160.144.191 TCP 54 36346 -> 443 [ACK] Seq=221 Ack=3021 Win=65535 Len=0
  8 0.040348121 192.168.35.166 -> 34.160.144.191 TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
  9 0.040581075 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=3021 Ack=314 Win=64240 Len=0
 10 0.054756955 34.160.144.191 -> 192.168.35.166 TLSv1.2 434 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
 11 0.102256611 192.168.35.166 -> 34.160.144.191 TCP 54 36346 -> 443 [ACK] Seq=314 Ack=3401 Win=65535 Len=0
 12 0.112673740 192.168.35.166 -> 34.160.144.191 TLSv1.2 153 Application Data
 13 0.112977736 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=3401 Ack=413 Win=64240 Len=0
 14 0.113453882 192.168.35.166 -> 34.160.144.191 TLSv1.2 92 Application Data
 15 0.113605447 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=3401 Ack=451 Win=64240 Len=0
 16 0.124096496 34.160.144.191 -> 192.168.35.166 TLSv1.2 92 Application Data
 17 0.124122401 192.168.35.166 -> 34.160.144.191 TCP 54 36346 -> 443 [ACK] Seq=451 Ack=3439 Win=65535 Len=0
 18 0.147574063 192.168.35.166 -> 34.160.144.191 TLSv1.2 323 Application Data
 19 0.147861461 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=3439 Ack=720 Win=64240 Len=0
 20 0.148143196 192.168.35.166 -> 34.160.144.191 TLSv1.2 181 Application Data
 21 0.148279455 34.160.144.191 -> 192.168.35.166 TCP 60 443 -> 36346 [ACK] Seq=3439 Ack=847 Win=64240 Len=0
 22 0.148657835 192.168.35.166 -> 34.107.221.82 TCP 74 41768 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=67040073 TSecr=0 WS=128
 23 0.148628674 34.107.221.82 -> 192.168.35.166 TCP 60 80 -> 41768 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
 24 0.160502162 192.168.35.166 -> 34.107.221.82 TCP 54 41768 -> 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
 25 0.160649486 192.168.35.166 -> 34.107.221.82 HTTP 368 GET /canonical.html HTTP/1.1
 26 0.160800014 34.107.221.82 -> 192.168.35.166 TCP 60 80 -> 41768 [ACK] Seq=1 Ack=315 Win=64240 Len=0
 27 0.161387593 34.160.144.191 -> 192.168.35.166 TLSv1.2 5871 Application Data, Application Data, Application Data, Application Data
 28 0.161442274 192.168.35.166 -> 34.160.144.191 TCP 54 36346 -> 443 [ACK] Seq=847 Ack=9256 Win=65535 Len=0

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Filtre de capture

"tshark -i ens33 -f "udp port 53"" port DNS

```

root@debian:~# tshark -i ens33 -f "udp port 53"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'ens33'
  1 0.000000000 192.168.35.166 -> 192.168.35.2 DNS 95 Standard query 0xb041 A content-signature-2.cdn.mozilla.net
  2 0.000895684 192.168.35.166 -> 192.168.35.2 DNS 95 Standard query 0xb043 AAAA content-signature-2.cdn.mozilla.net
  3 0.013322035 192.168.35.2 -> 192.168.35.166 DNS 111 Standard query response 0xb041 A content-signature-2.cdn.mozilla.net A 34.160.144.191
  4 0.013322272 192.168.35.2 -> 192.168.35.166 DNS 123 Standard query response 0xb043 AAAA content-signature-2.cdn.mozilla.net AAAA 2608:1901:0:92a9::
  5 0.264641745 192.168.35.166 -> 192.168.35.2 DNS 84 Standard query 0xf791 A detectportal.firefox.com
  6 0.264787935 192.168.35.166 -> 192.168.35.2 DNS 84 Standard query 0x3a97 AAAA detectportal.firefox.com
  7 0.274700317 192.168.35.2 -> 192.168.35.166 DNS 207 Standard query response 0x3a97 AAAA detectportal.firefox.com CNAME detectportal.prod.mozaws.net CNAME prod.detectportal.prod.cloudops.mozgcp.net AAAA 2608:1901:0:38d7::
  8 0.274700838 192.168.35.2 -> 192.168.35.166 DNS 195 Standard query response 0xf791 A detectportal.firefox.com CNAME detectportal.prod.mozaws.net CNAME prod.detectportal.prod.cloudops.mozgcp.net A 34.107.221.82
  9 0.317080533 192.168.35.166 -> 192.168.35.2 DNS 87 Standard query 0xff96 A services.addons.mozilla.org
 10 0.317158543 192.168.35.166 -> 192.168.35.2 DNS 87 Standard query 0xff94 AAAA services.addons.mozilla.org
 11 0.320667521 192.168.35.2 -> 192.168.35.166 DNS 151 Standard query response 0xff96 A services.addons.mozilla.org A 151.101.1.91 A 151.101.129.91 A 151.101.193.91 A 151.101.65.91
 12 0.320667839 192.168.35.2 -> 192.168.35.166 DNS 199 Standard query response 0xff94 AAAA services.addons.mozilla.org AAAA 2a04:4e42::347 AAAA 2a04:4e42:400::347 AAAA 2a04:4e42:600::347 AAAA 2a04:4e42:200::347
 13 0.363985042 192.168.35.166 -> 192.168.35.2 DNS 75 Standard query 0x1887 A ads.mozilla.org
 14 0.364071139 192.168.35.166 -> 192.168.35.2 DNS 75 Standard query 0x8086 AAAA ads.mozilla.org
 15 0.372810439 192.168.35.2 -> 192.168.35.166 DNS 175 Standard query response 0x1887 A ads.mozilla.org CNAME mozilla.map.fastly.net A 151.101.129.91 A 151.101.65.91 A 151.101.193.91 A 151.101.1.91
 16 0.374754778 192.168.35.2 -> 192.168.35.166 DNS 223 Standard query response 0x8086 AAAA ads.mozilla.org CNAME mozilla.map.fastly.net AAAA 2a04:4e42:200::347 AAAA 2a04:4e42:600::347 AAAA 2a04:4e42:400::347
 17 0.375721200 192.168.35.166 -> 192.168.35.2 DNS 97 Standard query 0x2a03 A firefox.settings.services.mozilla.com
 18 0.375804337 192.168.35.166 -> 192.168.35.2 DNS 97 Standard query 0xf101 AAAA firefox.settings.services.mozilla.com
 19 0.378832712 192.168.35.2 -> 192.168.35.166 DNS 245 Standard query response 0xf101 AAAA firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net AAAA 2a04:4e42::347 AAAA 2a04:4e42:200::347 AAAA 2a04:4e42:600::347
 20 0.378832712 192.168.35.2 -> 192.168.35.166 DNS 245 Standard query response 0xf101 AAAA firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net AAAA 2a04:4e42::347 AAAA 2a04:4e42:200::347 AAAA 2a04:4e42:600::347

```

Via Redirection (grep)

"tshark -i eth0 -f "tcp port 443" | grep "TLSv1.2|TLSv1.3""

```

root@debian:~# tshark -i ens33 -f "tcp port 443" | grep "TLSv1.2|TLSv1.3"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'ens33'
16 4 0.000702602 192.168.35.166 -> 34.160.144.191 TLSv1 274 Client Hello (SNI=content-signature-2.cdn.mozilla.net)
 6 0.037219728 34.160.144.191 -> 192.168.35.166 TLSv1.2 3074 Server Hello, Certificate, Server Key Exchange, Server Hello Done
 8 0.038760167 192.168.35.166 -> 34.160.144.191 TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
10 0.046725335 34.160.144.191 -> 192.168.35.166 TLSv1.2 434 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
11 0.047211808 192.168.35.166 -> 34.160.144.191 TLSv1.2 153 Application Data
12 0.047431685 192.168.35.166 -> 34.160.144.191 TLSv1.2 92 Application Data
15 0.060680982 34.160.144.191 -> 192.168.35.166 TLSv1.2 92 Application Data
20 0.324531409 192.168.35.166 -> 151.101.129.91 TLSv1 266 Client Hello (SNI=services.addons.mozilla.org)
22 0.331758884 151.101.129.91 -> 192.168.35.166 TLSv1.2 3070 Server Hello, Certificate, Server Key Exchange, Server Hello Done
24 0.332793709 192.168.35.166 -> 151.101.129.91 TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
26 0.335244651 151.101.129.91 -> 192.168.35.166 TLSv1.2 312 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
27 0.335575215 192.168.35.166 -> 151.101.129.91 TLSv1.2 153 Application Data
29 0.337901504 151.101.129.91 -> 192.168.35.166 TLSv1.2 126 Application Data
30 0.338016017 192.168.35.166 -> 151.101.129.91 TLSv1.2 92 Application Data
32 0.359827625 192.168.35.166 -> 151.101.129.91 TLSv1.2 400 Application Data
35 0.365702212 151.101.129.91 -> 192.168.35.166 TLSv1.2 1429 Application Data, Application Data
38 0.367593001 192.168.35.166 -> 151.101.1.91 TLSv1.2 1954 Client Hello (SNI=ads.mozilla.org)
44 0.372360329 192.168.35.166 -> 151.101.193.91 TLSv1 276 Client Hello (SNI=firefox.settings.services.mozilla.com)
46 0.373804007 151.101.1.91 -> 192.168.35.166 TLSv1.3 4583 Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, Application Data
48 0.376410157 151.101.193.91 -> 192.168.35.166 TLSv1.2 3114 Server Hello, Certificate, Server Key Exchange, Server Hello Done
50 0.381456311 192.168.35.166 -> 151.101.193.91 TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
52 0.382246048 192.168.35.166 -> 151.101.1.91 TLSv1.3 118 Change Cipher Spec, Application Data
56 0.383170580 192.168.35.166 -> 151.101.1.91 TLSv1.3 146 Application Data
58 0.383553116 192.168.35.166 -> 151.101.1.91 TLSv1.3 275 Application Data
60 0.383739518 192.168.35.166 -> 151.101.1.91 TLSv1.3 310 Application Data
62 0.384376848 151.101.193.91 -> 192.168.35.166 TLSv1.2 312 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
64 0.380458768 151.101.1.91 -> 192.168.35.166 TLSv1.3 119 Application Data
66 0.380612074 192.168.35.166 -> 151.101.1.91 TLSv1.3 85 Application Data
68 0.453515337 151.101.1.91 -> 192.168.35.166 TLSv1.3 12237 Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application Data
84 73 0.476034719 192.168.35.166 -> 151.101.193.91 TLSv1 276 Client Hello (SNI=firefox.settings.services.mozilla.com)

```